

TYPE	BANK	B23B Package
HSIO	2A_B	48
HSIO	2A_T	48
HPS shared HSIO	3A_B	48
HPS and SDM shared HSIO	3A_T	48
HVIO	6A	20
HVIO	6B	20
HVIO	6C	20
HVIO	6D	19
HVIO	6E	20
HVIO	6F	20
HVIO	6G	20
HVIO	6H	20
HPS I/O	HPS	48
SDM I/O	SDM	33

- i. Total LVDS channels per bank supporting SERDES Non-DPA and DPA mode is equivalent to (LVDS I/O per bank)/2, inclusive of clock pair. Please refer to Dedicated Tx/Rx Channel column in the pin-out table for the channel availability.
- ii. Total LVDS channels supporting SERDES Soft-CDR mode is 12 pairs per bank. Please refer to Soft CDR column in the pin out table for the channel availability.

Bank Number	Index within I/O Bank	Pin Name/Function	Optional Function(s)	Configuration Function	HPS Function	Dedicated Tx/Rx Channel SERDES	Soft CDR Support	B238	DQS for X4	DQS for X8/X9	DQS for X16/X18
SDM		TDO						BY67			
SDM		TMS						CY67			
SDM		TCK						CC84			
SDM		TDI						CC87			
SDM		OSC_CLK_1						DB72			
SDM		SDM_IO1	AVSTx8_DATA2,AS_DATA1					CK67			
SDM		SDM_IO5	AS_nCS00,MSEL0					BY71			
SDM		SDM_IO3	AVSTx8_DATA3,AS_DATA2					DC74			
SDM		nCONFIG						BK71			
SDM		SDM_IO4	AVSTx8_DATA1,AS_DATA0					CN64			
SDM		SDM_IO2	AVSTx8_DATA0,AS_CLK					CY69			
SDM		SDM_IO7	AS_nCS02,MSEL1					CN67			
SDM		SDM_IO11	AVSTx8_VALID,PWRMG_T_SDA					DG71			
SDM		nSTATUS						BM71			
SDM		SDM_IO16	PWRMG_T_SDA					DA74			
SDM		SDM_IO0	PWRMG_T_SCL,PWRMG_T_ALERT					BK67			
SDM		SDM_IO13	AVSTx8_DATA5					DA75			
SDM		SDM_IO9	AS_nCS01,MSEL2,PWRMG_T_ALERT					CN71			
SDM		SDM_IO6	AVSTx8_DATA4,AS_DATA3					DC75			
SDM		SDM_IO10	AVSTx8_DATA7					CK69			
SDM		SDM_IO8	AVSTx8_READY,AS_nCS03					DB71			
SDM		SDM_IO12	PWRMG_T_SDA,PWRMG_T_ALERT					BK69			
SDM		SDM_IO15	AVSTx8_DATA6,AS_nRST					CY71			
SDM		SDM_IO14	AVSTx8_CLK,PWRMG_T_SCL					BY69			
SDM		RREF_SDM						DH75			
SDM			TEMPDIODED0An					CN72			
SDM			TEMPDIODED0Ap					CK71			
SDM			VREFP_ADC					CF75			
SDM			VREFN_ADC					CF74			
SDM		VSIGP_0						BW75			
SDM		VSIGN_0						BW74			
SDM		VSIGP_1						CI74			
SDM		VSIGN_1						CI75			
HPS		HPS_IOA_1	GPIO0_IO0,SPIM0_SS1,N,SPIS0_CLK,UART0_CTS,N,NAND_ADQ0,SDMMC_DATA0,USB0_CLK,EMAC0_PPS0,TRACE_D10		HPS_IOA_1			BC64			
HPS		HPS_IOA_2	GPIO0_IO1,SPIM1_SS1,N,SPIS0_MOSI,UART0_RTS,N,NAND_ADQ1,SDMMC_DATA1,USB0_STP,EMAC0_PPSTRIG0,TRACE_D9		HPS_IOA_2			BC67			
HPS		HPS_IOA_3	GPIO0_IO2,SPIS0_SS0,N,UART0_TX,I2C1_SDA,NAND_WE,N,SDMMC_CLK,USB0_DIR,EMAC1_PPS1,TRACE_D8		HPS_IOA_3			AV67			
HPS		HPS_IOA_4	GPIO0_IO3,SPIS0_MISO,UART1_RX,I2C1_SCL,NAND_RE,N,USB0_DATA0,EMAC1_PPSTRIG1,TRACE_D7		HPS_IOA_4			AN72			
HPS		HPS_IOA_5	GPIO0_IO4,SPIM0_CLK,UART1_CTS,N,I2C0_SDA,NAND_WP,N,SDMMC_WRITE,PROTECT,USB0_DATA1,EMAC2_PPS2,TRACE_D6		HPS_IOA_5			AV69			
HPS		HPS_IOA_6	GPIO0_IO5,SPIM0_MOSI,UART1_RTS,N,I2C0_SCL,NAND_ADQ2,SDMMC_DATA2,USB0_NXT,EMAC2_PPSTRIG2,TRACE_D5		HPS_IOA_6			BA75			
HPS		HPS_IOA_7	GPIO0_IO6,SPIM0_MISO,MDIO2_MDI0,UART1_TX,I2C_EMAC2_SDA,NAND_ADQ3,SDMMC_DATA3,USB0_DATA2,TRACE_D4		HPS_IOA_7			BC71			
HPS		HPS_IOA_8	GPIO0_IO7,SPIM0_SS0,N,MDIO2_MDC,UART1_RX,I2C_EMAC2_SCL,NAND_CLE,SDMMC_CMD,USB0_DATA3,TRACE_D15		HPS_IOA_8			AJ74			
HPS		HPS_IOA_9	GPIO0_IO8,SPIM1_CLK,SPIS1_CLK,MDIO1_MDI0,I2C_EMAC1_SDA,NAND_ADQ4,SDMMC_DATA4,USB0_DATA4,I3C1_SDA,TRACE_D14		HPS_IOA_9			AY71			
HPS		HPS_IOA_10	GPIO0_IO9,SPIM1_MOSI,SPIS1_MOSI,MDIO1_MDC,I2C_EMAC1_SCL,NAND_ADQ5,SDMMC_DATA5,USB0_DATA5,I3C1_SCL,TRACE_D13		HPS_IOA_10			AJ75			
HPS		HPS_IOA_11	GPIO0_IO10,SPIM1_MISO,SPIS1_SS0,N,MDIO0_MDI0,I2C_EMAC0_SDA,NAND_ADQ6,SDMMC_DATA6,USB0_DATA6,I3C0_SDA,TRACE_D12		HPS_IOA_11			BC72			
HPS		HPS_IOA_12	GPIO0_IO11,SPIM1_SS0,N,SPIS1_MISO,MDIO0_MDC,I2C_EMAC0_SCL,NAND_ADQ7,SDMMC_DATA7,USB0_DATA7,I3C0_SCL,TRACE_D11		HPS_IOA_12			BP74			
HPS		HPS_IOA_13	GPIO0_IO12,NAND_ALE,SDMMC_PU_PD_DATA2,USB1_CLK,EMAC0_TX_CLK,TRACE_D10		HPS_IOA_13			BP75			
HPS		HPS_IOA_14	GPIO0_IO13,NAND_RB,N,SDMMC_PWR_ENA,USB1_STP,EMAC0_TX_CTL,TRACE_D9		HPS_IOA_14			BL74			
HPS		HPS_IOA_15	GPIO0_IO14,NAND_CE,N,USB1_DIR,EMAC0_RX_CLK,TRACE_D8		HPS_IOA_15			BL75			
HPS		HPS_IOA_16	GPIO0_IO15,NAND_DQS,SDMMC_DATA_STROBE,USB1_DATA0,EMAC0_RX_CTL,TRACE_D7		HPS_IOA_16			AP74			
HPS		HPS_IOA_17	GPIO0_IO16,I3C1_SDA,NAND_ADQ8,USB1_DATA1,EMAC0_TXD0,TRACE_D6		HPS_IOA_17			BG74			
HPS		HPS_IOA_18	GPIO0_IO17,I3C1_SCL,NAND_ADQ9,USB1_NXT,EMAC0_TXD1,TRACE_D5		HPS_IOA_18			AP75			
HPS		HPS_IOA_19	GPIO0_IO18,I3C0_SDA,NAND_ADQ10,USB1_DATA2,EMAC0_RXD0,TRACE_D4		HPS_IOA_19			BD74			
HPS		HPS_IOA_20	GPIO0_IO19,SPIM1_SS1,N,I3C0_SCL,NAND_ADQ11,USB1_DATA3,EMAC0_RXD1,TRACE_CLK		HPS_IOA_20			AN71			
HPS		HPS_IOA_21	GPIO0_IO20,SPIM1_CLK,SPIS0_CLK,UART0_CTS,N,I2C1_SDA,NAND_ADQ12,USB1_DATA4,EMAC0_TXD2,TRACE_D0		HPS_IOA_21			BD75			
HPS		HPS_IOA_22	GPIO0_IO21,SPIM1_MOSI,SPIS0_MOSI,UART0_RTS,N,I2C1_SCL,NAND_ADQ13,USB1_DATA5,EMAC0_TXD3,TRACE_D1		HPS_IOA_22			AM74			
HPS		HPS_IOA_23	GPIO0_IO22,SPIM1_MISO,SPIS0_SS0,N,UART0_TX,I2C0_SDA,NAND_ADQ14,USB1_DATA6,EMAC0_RXD2,TRACE_D2		HPS_IOA_23			AJ74			
HPS		HPS_IOA_24	GPIO0_IO23,SPIM1_SS0,N,SPIS0_MISO,UART0_RX,I2C0_SCL,NAND_ADQ15,USB1_DATA7,EMAC0_RXD3,TRACE_D3		HPS_IOA_24			AJ75			
HPS		HPS_IOB_1	GPIO1_IO0,SPIM1_CLK,UART0_CTS,N,EMAC0_PPS0,NAND_ADQ0,SDMMC_DATA0,I3C0_SDA,PULLUP_EN,EMAC1_TX_CLK,TRACE_D10		HPS_IOB_1			AF75			
HPS		HPS_IOB_2	GPIO1_IO1,SPIM1_MOSI,UART0_RTS,N,EMAC0_PPSTRIG0,NAND_ADQ1,SDMMC_DATA1,I3C1_SDA,PULLUP_EN,EMAC1_TX_CTL,TRACE_D9		HPS_IOB_2			AC75			
HPS		HPS_IOB_3	GPIO1_IO2,SPIM1_MISO,UART0_TX,I2C0_SDA,NAND_WE,N,SDMMC_CLK,EMAC1_RX_CLK,TRACE_D8		HPS_IOB_3			AC74			
HPS		HPS_IOB_4	GPIO1_IO3,SPIM1_SS0,N,UART0_RX,I2C0_SCL,NAND_RE,N,EMAC1_RX_CTL,TRACE_D7		HPS_IOB_4			AN67			
HPS		HPS_IOB_5	GPIO1_IO4,SPIM1_SS1,N,SPIS1_CLK,UART1_CTS,N,EMAC2_PPS2,NAND_WP,N,SDMMC_WRITE,PROTECT,I3C1_SDA,EMAC1_TXD0,TRACE_D6		HPS_IOB_5			Y75			
HPS		HPS_IOB_6	GPIO1_IO5,SPIS1_MOSI,UART1_RTS,N,EMAC2_PPSTRIG2,NAND_ADQ2,SDMMC_DATA2,I3C1_SCL,EMAC1_TXD1,TRACE_D5		HPS_IOB_6			AN64			
HPS		HPS_IOB_7	GPIO1_IO6,SPIS1_SS0,N,UART1_TX,I2C1_SDA,NAND_ADQ3,SDMMC_DATA3,I3C0_SDA,EMAC1_RXD0,TRACE_D4		HPS_IOB_7			Y74			
HPS		HPS_IOB_8	GPIO1_IO7,SPIS1_MISO,UART1_RX,I2C1_SCL,NAND_CLE,SDMMC_CMD,I3C0_SCL,EMAC1_RXD1,TRACE_D15		HPS_IOB_8			AK69			
HPS		HPS_IOB_9	GPIO1_IO8,JTAG_TCK,SPIS0_CLK,MDIO2_MDI0,I2C_EMAC2_SDA,NAND_ADQ4,SDMMC_DATA4,EMAC1_TXD2,TRACE_D14		HPS_IOB_9			U74			
HPS		HPS_IOB_10	GPIO1_IO9,JTAG_TMS,SPIS0_MOSI,MDIO2_MDC,I2C_EMAC2_SCL,NAND_ADQ5,SDMMC_DATA5,EMAC1_TXD3,TRACE_D13		HPS_IOB_10			AK67			
HPS		HPS_IOB_11	GPIO1_IO10,JTAG_TDO,SPIS0_SS0,N,MDIO0_MDI0,I2C_EMAC0_SDA,NAND_ADQ6,SDMMC_DATA6,EMAC1_RXD2,TRACE_D12		HPS_IOB_11			P75			
HPS		HPS_IOB_12	GPIO1_IO11,JTAG_TDI,SPIS0_MISO,MDIO0_MDC,I2C_EMAC0_SCL,NAND_ADQ7,SDMMC_DATA7,EMAC1_RXD3,TRACE_D11		HPS_IOB_12			P74			
HPS		HPS_IOB_13	GPIO1_IO12,I2C1_SDA,NAND_ALE,SDMMC_PU_PD_DATA2,EMAC2_TX_CLK,TRACE_D10		HPS_IOB_13			L75			
HPS		HPS_IOB_14	GPIO1_IO13,I2C1_SCL,NAND_RB,N,SDMMC_PWR_ENA,EMAC2_TX_CTL,TRACE_D9		HPS_IOB_14			N72			
HPS		HPS_IOB_15	GPIO1_IO14,UART1_TX,NAND_CE,N,I3C1_SDA,EMAC2_RX_CLK,TRACE_D8		HPS_IOB_15			N71			
HPS		HPS_IOB_16	GPIO1_IO15,UART1_RX,NAND_DQS,SDMMC_DATA_STROBE,I3C1_SCL,EMAC2_RX_CTL,TRACE_D7		HPS_IOB_16			AD72			
HPS		HPS_IOB_17	GPIO1_IO16,UART1_CTS,N,NAND_ADQ8,I3C0_SDA,EMAC2_TXD0,TRACE_D6		HPS_IOB_17			F75			
HPS		HPS_IOB_18	GPIO1_IO17,SPIM0_SS1,N,UART1_RTS,N,NAND_ADQ9,I3C0_SCL,EMAC2_TXD1,TRACE_D5		HPS_IOB_18			AD71			
HPS		HPS_IOB_19	GPIO1_IO18,SPIM0_MISO,MDIO1_MDI0,I2C_EMAC1_SDA,NAND_ADQ10,EMAC2_RXD0,TRACE_D4		HPS_IOB_19			K71			
HPS		HPS_IOB_20	GPIO1_IO19,SPIM0_SS0,N,MDIO1_MDC,I2C_EMAC1_SCL,NAND_ADQ11,EMAC2_RXD1,TRACE_CLK		HPS_IOB_20			AK71			
HPS		HPS_IOB_21	GPIO1_IO20,SPIM0_CLK,SPIS1_CLK,I2C_EMAC2_SDA,NAND_ADQ12,I3C0_SDA,PULLUP_EN,EMAC2_TXD2,TRACE_D0		HPS_IOB_21			F74			
HPS		HPS_IOB_22	GPIO1_IO21,SPIM0_MOSI,SPIS1_MOSI,I2C_EMAC2_SCL,NAND_ADQ13,I3C1_SDA,PULLUP_EN,EMAC2_TXD3,TRACE_D1		HPS_IOB_22			AA71			

Bank Number	Index within I/O Bank	Pin Name/Function	Optional Function(s)	Configuration Function	HPS Function	Dedicated Tx/Rx Channel SERDES	Soft CDR Support	B238	DQS for X4	DQS for X8/X9	DQS for X16/X18
HPS		HPS_I0B_23	GPIO1_I022,SPIM0_MISO,SPIS1_SS0_N,MDIO0_MDIO,I2C_EMAC0_SDA,NAND_ADQ14,EMAC2_RXD2,TRACE_D2		HPS_I0B_23			C74			
HPS		HPS_I0B_24	GPIO1_I023,SPIM0_SS0_N,SPIS1_MISO,MDIO0_MDC,I2C_EMAC0_SCL,NAND_ADQ15,EMAC2_RXD3,TRACE_D3		HPS_I0B_24			D71			
2A_T	95	I0	I0B			DIFF_I0_2A_T1n	No	DJ9	DQ0	DQ0	DQ0
2A_T	94	I0	I0B			DIFF_I0_2A_T1p	No	DK9	DQ0	DQ0	DQ0
2A_T	93	I0	I0B			DIFF_I0_2A_T2n	No	DO9	DQ0	DQ0	DQ0
2A_T	92	I0	I0B			DIFF_I0_2A_T2p	No	DD12	DQ0	DQ0	DQ0
2A_T	91	I0	I0B			DIFF_I0_2A_T3n	No	DF12	DQSn0	DQ0	DQ0
2A_T	90	I0	I0B			DIFF_I0_2A_T3p	No	DJ12	DQS0	DQ0	DQ0
2A_T	89	I0	I0B			DIFF_I0_2A_T4n	No	DJ18	DQSn1	DQSn0/CQn0	DQ0
2A_T	88	I0	I0B			DIFF_I0_2A_T4p	No	DK18	DQS1	DQS0/CQ0	DQ0
2A_T	87	I0	I0B			DIFF_I0_2A_T5n	Yes	DD18	DQ1	DQ0	DQ0
2A_T	86	I0	I0B			DIFF_I0_2A_T5p	Yes	DD21	DQ1	DQ0	DQ0
2A_T	85	I0	I0B			DIFF_I0_2A_T6n	No	DF21	DQ1	DQ0	DQ0
2A_T	84	I0	I0B			DIFF_I0_2A_T6p	No	DJ21	DQ1	DQ0	DQ0
2A_T	83	I0	I0B			DIFF_I0_2A_T7n	No	DN7	DQ2	DQ1	DQ0
2A_T	82	I0	I0B			DIFF_I0_2A_T7p	No	DN3	DQ2	DQ1	DQ0
2A_T	81	I0	I0B			DIFF_I0_2A_T8n	No	DN10	DQ2	DQ1	DQ0
2A_T	80	I0	I0B			DIFF_I0_2A_T8p	No	DP7	DQ2	DQ1	DQ0
2A_T	79	I0	I0B			DIFF_I0_2A_T9n	No	DP13	DQSn2	DQ1	DQ0
2A_T	78	I0	I0B			DIFF_I0_2A_T9p	No	DP10	DQS2	DQ1	DQ0
2A_T	77	I0	I0B			DIFF_I0_2A_T10n	No	DN15	DQSn3	DQSn1/CQn1	DQSn0/CQn0
2A_T	76	I0	I0B			DIFF_I0_2A_T10p	No	DP15	DQS3	DQS1/CQ1	DQS0/CQ0
2A_T	75	I0	I0B			DIFF_I0_2A_T11n	Yes	DN20	DQ3	DQ1	DQ0
2A_T	74	I0	I0B			DIFF_I0_2A_T11p	Yes	DP22	DQ3	DQ1	DQ0
2A_T	73	I0	I0B			DIFF_I0_2A_T12n	No	DN17	DQ3	DQ1	DQ0
2A_T	72	I0	I0B			DIFF_I0_2A_T12p	No	DP20	DQ3	DQ1	DQ0
2A_T	71	I0	I0B			DIFF_I0_2A_T13n	No	DJ24	DQ4	DQ2	DQ1
2A_T	70	I0	I0B			DIFF_I0_2A_T13p	No	DK24	DQ4	DQ2	DQ1
2A_T	69	I0	I0B			DIFF_I0_2A_T14n	No	DD24	DQ4	DQ2	DQ1
2A_T	68	I0	I0B			DIFF_I0_2A_T14p	No	DD27	DQ4	DQ2	DQ1
2A_T	67	I0	I0B			DIFF_I0_2A_T15n	No	DF27	DQSn4	DQ2	DQ1
2A_T	66	I0	I0B			DIFF_I0_2A_T15p	No	DJ27	DQS4	DQ2	DQ1
2A_T	65	I0	I0B,PLL_2A_T_CLKOUT1n			DIFF_I0_2A_T16n	No	DJ32	DQSn5	DQSn2/CQn2	DQ1
2A_T	64	I0	I0B,PLL_2A_T_CLKOUT1p,PLL_2A_T_CLKOUT1,PLL_2A_T_FB1			DIFF_I0_2A_T16p	No	DK32	DQS5	DQS2/CQ2	DQ1
2A_T	63	I0	I0B			DIFF_I0_2A_T17n	Yes	DD32	DQ5	DQ2	DQ1
2A_T	62	I0	I0B,RZQ_T_2A			DIFF_I0_2A_T17p	Yes	DD35	DQ5	DQ2	DQ1
2A_T	61	I0	I0B,CLK_T_2A_1n			DIFF_I0_2A_T18n	No	DF35	DQ5	DQ2	DQ1
2A_T	60	I0	I0B,CLK_T_2A_1p			DIFF_I0_2A_T18p	No	DJ35	DQ5	DQ2	DQ1
2A_T	59	I0	I0B,CLK_T_2A_0n			DIFF_I0_2A_T19n	No	DN22	DQ6	DQ3	DQ1
2A_T	58	I0	I0B,CLK_T_2A_0p			DIFF_I0_2A_T19p	No	DP23	DQ6	DQ3	DQ1
2A_T	57	I0	I0B			DIFF_I0_2A_T20n	No	DN25	DQ6	DQ3	DQ1
2A_T	56	I0	I0B			DIFF_I0_2A_T20p	No	DP25	DQ6	DQ3	DQ1
2A_T	55	I0	I0B,PLL_2A_T_CLKOUT0n			DIFF_I0_2A_T21n	No	DN28	DQSn6	DQ3	DQ1
2A_T	54	I0	I0B,PLL_2A_T_CLKOUT0p,PLL_2A_T_CLKOUT0,PLL_2A_T_FB0			DIFF_I0_2A_T21p	No	DP30	DQS6	DQ3	DQ1
2A_T	53	I0	I0B			DIFF_I0_2A_T22n	No	DN30	DQSn7	DQSn3/CQn3	DQSn1/CQn1
2A_T	52	I0	I0B			DIFF_I0_2A_T22p	No	DP33	DQS7	DQS3/CQ3	DQS1/CQ1
2A_T	51	I0	I0B			DIFF_I0_2A_T23n	Yes	DN38	DQ7	DQ3	DQ1
2A_T	50	I0	I0B			DIFF_I0_2A_T23p	Yes	DP38	DQ7	DQ3	DQ1
2A_T	49	I0	I0B			DIFF_I0_2A_T24n	No	DN33	DQ7	DQ3	DQ1
2A_T	48	I0	I0B			DIFF_I0_2A_T24p	No	DP36	DQ7	DQ3	DQ1
2A_B	47	I0	I0B			DIFF_I0_2A_B1n	No	DN41	DQ8	DQ4	DQ2
2A_B	46	I0	I0B			DIFF_I0_2A_B1p	No	DP43	DQ8	DQ4	DQ2
2A_B	45	I0	I0B			DIFF_I0_2A_B2n	No	DN43	DQ8	DQ4	DQ2
2A_B	44	I0	I0B			DIFF_I0_2A_B2p	No	DP45	DQ8	DQ4	DQ2
2A_B	43	I0	I0B			DIFF_I0_2A_B3n	No	DN45	DQSn8	DQ4	DQ2
2A_B	42	I0	I0B			DIFF_I0_2A_B3p	No	DP47	DQS8	DQ4	DQ2
2A_B	41	I0	I0B,PLL_2A_B_CLKOUT1n			DIFF_I0_2A_B4n	No	DN50	DQSn9	DQSn4/CQn4	DQ2
2A_B	40	I0	I0B,PLL_2A_B_CLKOUT1p,PLL_2A_B_CLKOUT1,PLL_2A_B_FB1			DIFF_I0_2A_B4p	No	DP50	DQS9	DQS4/CQ4	DQ2
2A_B	39	I0	I0B			DIFF_I0_2A_B5n	Yes	DN55	DQ9	DQ4	DQ2
2A_B	38	I0	I0B,RZQ_B_2A			DIFF_I0_2A_B5p	Yes	DP58	DQ9	DQ4	DQ2
2A_B	37	I0	I0B,CLK_B_2A_1n			DIFF_I0_2A_B6n	Yes	DN53	DQ9	DQ4	DQ2
2A_B	36	I0	I0B,CLK_B_2A_1p			DIFF_I0_2A_B6p	Yes	DP55	DQ9	DQ4	DQ2
2A_B	35	I0	I0B,CLK_B_2A_0n			DIFF_I0_2A_B7n	No	DJ39	DQ10	DQ5	DQ2
2A_B	34	I0	I0B,CLK_B_2A_0p			DIFF_I0_2A_B7p	No	DK39	DQ10	DQ5	DQ2
2A_B	33	I0	I0B			DIFF_I0_2A_B8n	No	DD39	DQ10	DQ5	DQ2
2A_B	32	I0	I0B			DIFF_I0_2A_B8p	No	DD42	DQ10	DQ5	DQ2
2A_B	31	I0	I0B,PLL_2A_B_CLKOUT0n			DIFF_I0_2A_B9n	No	DF42	DQSn10	DQ5	DQ2
2A_B	30	I0	I0B,PLL_2A_B_CLKOUT0p,PLL_2A_B_CLKOUT0,PLL_2A_B_FB0			DIFF_I0_2A_B9p	No	DJ42	DQS10	DQ5	DQ2
2A_B	29	I0	I0B			DIFF_I0_2A_B10n	No	DJ48	DQSn11	DQSn5/CQn5	DQSn2/CQn2
2A_B	28	I0	I0B			DIFF_I0_2A_B10p	No	DK48	DQS11	DQS5/CQ5	DQS2/CQ2
2A_B	27	I0	I0B			DIFF_I0_2A_B11n	Yes	DD48	DQ11	DQ5	DQ2
2A_B	26	I0	I0B			DIFF_I0_2A_B11p	Yes	DD51	DQ11	DQ5	DQ2
2A_B	25	I0	I0B			DIFF_I0_2A_B12n	Yes	DF51	DQ11	DQ5	DQ2
2A_B	24	I0	I0B			DIFF_I0_2A_B12p	Yes	DJ51	DQ11	DQ5	DQ2
2A_B	23	I0	I0B			DIFF_I0_2A_B13n	No	DN58	DQ12	DQ6	DQ3
2A_B	22	I0	I0B			DIFF_I0_2A_B13p	No	DP60	DQ12	DQ6	DQ3
2A_B	21	I0	I0B			DIFF_I0_2A_B14n	No	DN61	DQ12	DQ6	DQ3
2A_B	20	I0	I0B			DIFF_I0_2A_B14p	No	DP61	DQ12	DQ6	DQ3
2A_B	19	I0	I0B			DIFF_I0_2A_B15n	No	DN63	DQSn12	DQ6	DQ3

Bank Number	Index within I/O Bank	Pin Name/Function	Optional Function(s)	Configuration Function	HPS Function	Dedicated Tx/Rx Channel SERDES	Soft CDR Support	B238	DQS for X4	DQS for X8/X9	DQS for X16/X18
2A_B	18	IO	IOB			DIFF_IO_2A_B15p	No	DP66	DQS12	DQ6	DQ3
2A_B	17	IO	IOB			DIFF_IO_2A_B16n	No	DN66	DQSn13	DQSn6/CQn6	DQ3
2A_B	16	IO	IOB			DIFF_IO_2A_B16p	No	DP68	DQS13	DQS6/CQ6	DQ3
2A_B	15	IO	IOB			DIFF_IO_2A_B17n	Yes	DN74	DQ13	DQ6	DQ3
2A_B	14	IO	IOB			DIFF_IO_2A_B17p	Yes	DN73	DQ13	DQ6	DQ3
2A_B	13	IO	IOB			DIFF_IO_2A_B18n	Yes	DN68	DQ13	DQ6	DQ3
2A_B	12	IO	IOB			DIFF_IO_2A_B18p	Yes	DP70	DQ13	DQ6	DQ3
2A_B	11	IO	IOB			DIFF_IO_2A_B19n	No	DJ56	DQ14	DQ7	DQ3
2A_B	10	IO	IOB			DIFF_IO_2A_B19p	No	DK56	DQ14	DQ7	DQ3
2A_B	9	IO	IOB			DIFF_IO_2A_B20n	No	DD56	DQ14	DQ7	DQ3
2A_B	8	IO	IOB			DIFF_IO_2A_B20p	No	DD59	DQ14	DQ7	DQ3
2A_B	7	IO	IOB			DIFF_IO_2A_B21n	No	DF59	DQSn14	DQ7	DQ3
2A_B	6	IO	IOB			DIFF_IO_2A_B21p	No	DJ59	DQS14	DQ7	DQ3
2A_B	5	IO	IOB			DIFF_IO_2A_B22n	No	DJ62	DQSn15	DQSn7/CQn7	DQS3/CQn3
2A_B	4	IO	IOB			DIFF_IO_2A_B22p	No	DK62	DQS15	DQS7/CQ7	DQS3/CQ3
2A_B	3	IO	IOB			DIFF_IO_2A_B23n	Yes	DD62	DQ15	DQ7	DQ3
2A_B	2	IO	IOB			DIFF_IO_2A_B23p	Yes	DD65	DQ15	DQ7	DQ3
2A_B	1	IO	IOB			DIFF_IO_2A_B24n	Yes	DF65	DQ15	DQ7	DQ3
2A_B	0	IO	IOB			DIFF_IO_2A_B24p	Yes	DJ65	DQ15	DQ7	DQ3
3A_T	95	IO	IOB		HPS_DDR	DIFF_IO_3A_T1n	No	A20	DQ16	DQ8	DQ4
3A_T	94	IO	IOB		HPS_DDR	DIFF_IO_3A_T1p	No	B22	DQ16	DQ8	DQ4
3A_T	93	IO	IOB		HPS_DDR	DIFF_IO_3A_T2n	No	A23	DQ16	DQ8	DQ4
3A_T	92	IO	IOB		HPS_DDR	DIFF_IO_3A_T2p	No	A22	DQ16	DQ8	DQ4
3A_T	91	IO	IOB		HPS_DDR	DIFF_IO_3A_T3n	No	B25	DQS16	DQ8	DQ4
3A_T	90	IO	IOB	AVST_READY	HPS_DDR	DIFF_IO_3A_T3p	No	A25	DQS16	DQ8	DQ4
3A_T	89	IO	IOB		HPS_DDR	DIFF_IO_3A_T4n	No	A30	DQS17	DQS8/CQn8	DQ4
3A_T	88	IO	IOB		HPS_DDR	DIFF_IO_3A_T4p	No	B28	DQS17	DQS8/CQ8	DQ4
3A_T	87	IO	IOB		HPS_DDR	DIFF_IO_3A_T5n	Yes	A33	DQ17	DQ8	DQ4
3A_T	86	IO	IOB		HPS_DDR	DIFF_IO_3A_T5p	Yes	B30	DQ17	DQ8	DQ4
3A_T	85	IO	IOB		HPS_DDR	DIFF_IO_3A_T6n	No	A36	DQ17	DQ8	DQ4
3A_T	84	IO	IOB		HPS_DDR	DIFF_IO_3A_T6p	No	B33	DQ17	DQ8	DQ4
3A_T	83	IO	IOB	AVST_DATA10	HPS_DDR	DIFF_IO_3A_T7n	No	E24	DQ18	DQ9	DQ4
3A_T	82	IO	IOB	AVST_DATA9	HPS_DDR	DIFF_IO_3A_T7p	No	G24	DQ18	DQ9	DQ4
3A_T	81	IO	IOB	AVST_DATA8	HPS_DDR	DIFF_IO_3A_T8n	No	M24	DQ18	DQ9	DQ4
3A_T	80	IO	IOB	AVST_VALID	HPS_DDR	DIFF_IO_3A_T8p	No	T24	DQ18	DQ9	DQ4
3A_T	79	IO	IOB	AVST_DATA7	HPS_DDR	DIFF_IO_3A_T9n	No	G27	DQS18	DQ9	DQ4
3A_T	78	IO	IOB	AVST_DATA6	HPS_DDR	DIFF_IO_3A_T9p	No	J27	DQS18	DQ9	DQ4
3A_T	77	IO	IOB	AVST_DATA5	HPS_DDR	DIFF_IO_3A_T10n	No	E32	DQS19	DQS9/CQn9	DQS4/CQn4
3A_T	76	IO	IOB	AVST_DATA4	HPS_DDR	DIFF_IO_3A_T10p	No	G32	DQS19	DQS9/CQ9	DQS4/CQ4
3A_T	75	IO	IOB	AVST_DATA3	HPS_DDR	DIFF_IO_3A_T11n	Yes	M32	DQ19	DQ9	DQ4
3A_T	74	IO	IOB	AVST_DATA2	HPS_DDR	DIFF_IO_3A_T11p	Yes	T32	DQ19	DQ9	DQ4
3A_T	73	IO	IOB	AVST_DATA1	HPS_DDR	DIFF_IO_3A_T12n	No	G35	DQ19	DQ9	DQ4
3A_T	72	IO	IOB	AVST_DATA0	HPS_DDR	DIFF_IO_3A_T12p	No	J35	DQ19	DQ9	DQ4
3A_T	71	IO	IOB		HPS_DDR	DIFF_IO_3A_T13n	No	M27	DQ20	DQ10	DQ5
3A_T	70	IO	IOB		HPS_DDR	DIFF_IO_3A_T13p	No	T27	DQ20	DQ10	DQ5
3A_T	69	IO	IOB		HPS_DDR	DIFF_IO_3A_T14n	No	W24	DQ20	DQ10	DQ5
3A_T	68	IO	IOB		HPS_DDR	DIFF_IO_3A_T14p	No	AB24	DQ20	DQ10	DQ5
3A_T	67	IO	IOB		HPS_DDR	DIFF_IO_3A_T15n	No	M35	DQS20	DQ10	DQ5
3A_T	66	IO	IOB		HPS_DDR	DIFF_IO_3A_T15p	No	T35	DQS20	DQ10	DQ5
3A_T	65	IO	IOB,PLL_3A_T_CLKOUT1n		HPS_DDR	DIFF_IO_3A_T16n	No	W32	DQS21	DQS10/CQn10	DQ5
3A_T	64	IO	IOB,PLL_3A_T_CLKOUT1p,PLL_3A_T_CLKOUT1,PLL_3A_T_FB1		HPS_DDR	DIFF_IO_3A_T16p	No	AB32	DQS21	DQS10/CQ10	DQ5
3A_T	63	IO	IOB		HPS_DDR	DIFF_IO_3A_T17n	Yes	T42	DQ21	DQ10	DQ5
3A_T	62	IO	IOB,RZQ_T_3A		HPS_DDR	DIFF_IO_3A_T17p	Yes	M42	DQ21	DQ10	DQ5
3A_T	61	IO	IOB,CLK_T_3A_1n		HPS_DDR	DIFF_IO_3A_T18n	No	W39	DQ21	DQ10	DQ5
3A_T	60	IO	IOB,CLK_T_3A_1p		HPS_DDR	DIFF_IO_3A_T18p	No	AB39	DQ21	DQ10	DQ5
3A_T	59	IO	IOB,CLK_T_3A_0n		HPS_DDR	DIFF_IO_3A_T19n	No	B38	DQ22	DQ11	DQ5
3A_T	58	IO	IOB,CLK_T_3A_0p		HPS_DDR	DIFF_IO_3A_T19p	No	A38	DQ22	DQ11	DQ5
3A_T	57	IO	IOB		HPS_DDR	DIFF_IO_3A_T20n	No	A43	DQ22	DQ11	DQ5
3A_T	56	IO	IOB		HPS_DDR	DIFF_IO_3A_T20p	No	B41	DQ22	DQ11	DQ5
3A_T	55	IO	IOB,PLL_3A_T_CLKOUT0n		HPS_DDR	DIFF_IO_3A_T21n	No	A45	DQS22	DQ11	DQ5
3A_T	54	IO	IOB,PLL_3A_T_CLKOUT0p,PLL_3A_T_CLKOUT0,PLL_3A_T_FB0		HPS_DDR	DIFF_IO_3A_T21p	No	B43	DQS22	DQ11	DQ5
3A_T	53	IO	IOB	AVST_CLK	HPS_DDR	DIFF_IO_3A_T22n	No	A47	DQS23	DQS11/CQn11	DQS5/CQn5
3A_T	52	IO	IOB	AVST_DATA15	HPS_DDR	DIFF_IO_3A_T22p	No	B45	DQS23	DQS11/CQ11	DQS5/CQ5
3A_T	51	IO	IOB	AVST_DATA14	HPS_DDR	DIFF_IO_3A_T23n	Yes	A55	DQ23	DQ11	DQ5
3A_T	50	IO	IOB	AVST_DATA13	HPS_DDR	DIFF_IO_3A_T23p	Yes	B53	DQ23	DQ11	DQ5
3A_T	49	IO	IOB	AVST_DATA12	HPS_DDR	DIFF_IO_3A_T24n	No	B50	DQ23	DQ11	DQ5
3A_T	48	IO	IOB	AVST_DATA11	HPS_DDR	DIFF_IO_3A_T24p	No	A50	DQ23	DQ11	DQ5
3A_B	47	IO	IOB		HPS_DDR	DIFF_IO_3A_B1n	No	E39	DQ24	DQ12	DQ6
3A_B	46	IO	IOB		HPS_DDR	DIFF_IO_3A_B1p	No	G39	DQ24	DQ12	DQ6
3A_B	45	IO	IOB		HPS_DDR	DIFF_IO_3A_B2n	No	M39	DQ24	DQ12	DQ6
3A_B	44	IO	IOB		HPS_DDR	DIFF_IO_3A_B2p	No	T39	DQ24	DQ12	DQ6
3A_B	43	IO	IOB		HPS_DDR	DIFF_IO_3A_B3n	No	G42	DQS24	DQ12	DQ6
3A_B	42	IO	IOB		HPS_DDR	DIFF_IO_3A_B3p	No	J42	DQS24	DQ12	DQ6
3A_B	41	IO	IOB,PLL_3A_B_CLKOUT1n		HPS_DDR	DIFF_IO_3A_B4n	No	E48	DQS25	DQS12/CQn12	DQ6
3A_B	40	IO	IOB,PLL_3A_B_CLKOUT1p,PLL_3A_B_CLKOUT1,PLL_3A_B_FB1		HPS_DDR	DIFF_IO_3A_B4p	No	G48	DQS25	DQS12/CQ12	DQ6
3A_B	39	IO	IOB		HPS_DDR	DIFF_IO_3A_B5n	Yes	M48	DQ25	DQ12	DQ6
3A_B	38	IO	IOB,RZQ_B_3A		HPS_DDR	DIFF_IO_3A_B5p	Yes	T48	DQ25	DQ12	DQ6
3A_B	37	IO	IOB,CLK_B_3A_1n		HPS_DDR	DIFF_IO_3A_B6n	Yes	G51	DQ25	DQ12	DQ6
3A_B	36	IO	IOB,CLK_B_3A_1p		HPS_DDR	DIFF_IO_3A_B6p	Yes	J51	DQ25	DQ12	DQ6

Bank Number	Index within I/O Bank	Pin Name/Function	Optional Function(s)	Configuration Function	HPS Function	Dedicated Tx/Rx Channel SERDES	Soft CDR Support	B238	DQS for X4	DQS for X8/X9	DQS for X16/X18
3A_B	35	IO	IOB_CLK_B_3A_0n		HPS_DDR	DIFF_IO_3A_87n	No	A58	DQ26	DQ13	DQ6
3A_B	34	IO	IOB_CLK_B_3A_0p		HPS_DDR	DIFF_IO_3A_87p	No	B55	DQ26	DQ13	DQ6
3A_B	33	IO	IOB		HPS_DDR	DIFF_IO_3A_88n	No	A60	DQ26	DQ13	DQ6
3A_B	32	IO	IOB		HPS_DDR	DIFF_IO_3A_88p	No	B58	DQ26	DQ13	DQ6
3A_B	31	IO	IOB_PLL_3A_B_CLKOUT0n		HPS_DDR	DIFF_IO_3A_89n	No	B61	DQSn26	DQ13	DQ6
3A_B	30	IO	IOB_PLL_3A_B_CLKOUT0p,PLL_3A_B_CLKOUT0,PLL_3A_B_FB0		HPS_DDR	DIFF_IO_3A_89p	No	A61	DQSn26	DQ13	DQ6
3A_B	29	IO	IOB		HPS_DDR	DIFF_IO_3A_810n	No	A66	DQSn27	DQSn13/CQn13	DQSn6/CQn6
3A_B	28	IO	IOB		HPS_DDR	DIFF_IO_3A_810p	No	B63	DQSn27	DQSn13/CQn13	DQSn6/CQn6
3A_B	27	IO	IOB		HPS_DDR	DIFF_IO_3A_811n	Yes	A70	DQ27	DQ13	DQ6
3A_B	26	IO	IOB		HPS_DDR	DIFF_IO_3A_811p	Yes	B68	DQ27	DQ13	DQ6
3A_B	25	IO	IOB		HPS_DDR	DIFF_IO_3A_812n	Yes	A68	DQ27	DQ13	DQ6
3A_B	24	IO	IOB		HPS_DDR	DIFF_IO_3A_812p	Yes	B66	DQ27	DQ13	DQ6
3A_B	23	IO	IOB		HPS_DDR	DIFF_IO_3A_813n	No	M51	DQ28	DQ14	DQ7
3A_B	22	IO	IOB		HPS_DDR	DIFF_IO_3A_813p	No	T51	DQ28	DQ14	DQ7
3A_B	21	IO	IOB		HPS_DDR	DIFF_IO_3A_814n	No	W48	DQ28	DQ14	DQ7
3A_B	20	IO	IOB		HPS_DDR	DIFF_IO_3A_814p	No	AB48	DQ28	DQ14	DQ7
3A_B	19	IO	IOB		HPS_DDR	DIFF_IO_3A_815n	No	M59	DQSn28	DQ14	DQ7
3A_B	18	IO	IOB		HPS_DDR	DIFF_IO_3A_815p	No	T59	DQSn28	DQ14	DQ7
3A_B	17	IO	IOB		HPS_DDR	DIFF_IO_3A_816n	No	W56	DQSn29	DQSn14/CQn14	DQ7
3A_B	16	IO	IOB		HPS_DDR	DIFF_IO_3A_816p	No	AB56	DQSn29	DQSn14/CQn14	DQ7
3A_B	15	IO	IOB		HPS_DDR	DIFF_IO_3A_817n	Yes	M65	DQ29	DQ14	DQ7
3A_B	14	IO	IOB		HPS_DDR	DIFF_IO_3A_817p	Yes	T65	DQ29	DQ14	DQ7
3A_B	13	IO	IOB		HPS_DDR	DIFF_IO_3A_818n	Yes	W62	DQ29	DQ14	DQ7
3A_B	12	IO	IOB		HPS_DDR	DIFF_IO_3A_818p	Yes	AB62	DQ29	DQ14	DQ7
3A_B	11	IO	IOB		HPS_DDR	DIFF_IO_3A_819n	No	E56	DQ30	DQ15	DQ7
3A_B	10	IO	IOB		HPS_DDR	DIFF_IO_3A_819p	No	G56	DQ30	DQ15	DQ7
3A_B	9	IO	IOB		HPS_DDR	DIFF_IO_3A_820n	No	M56	DQ30	DQ15	DQ7
3A_B	8	IO	IOB		HPS_DDR	DIFF_IO_3A_820p	No	T56	DQ30	DQ15	DQ7
3A_B	7	IO	IOB		HPS_DDR	DIFF_IO_3A_821n	No	G59	DQSn30	DQ15	DQ7
3A_B	6	IO	IOB		HPS_DDR	DIFF_IO_3A_821p	No	J59	DQSn30	DQ15	DQ7
3A_B	5	IO	IOB		HPS_DDR	DIFF_IO_3A_822n	No	E62	DQSn31	DQSn15/CQn15	DQSn7/CQn7
3A_B	4	IO	IOB		HPS_DDR	DIFF_IO_3A_822p	No	G62	DQSn31	DQSn15/CQn15	DQSn7/CQn7
3A_B	3	IO	IOB		HPS_DDR	DIFF_IO_3A_823n	Yes	M62	DQ31	DQ15	DQ7
3A_B	2	IO	IOB		HPS_DDR	DIFF_IO_3A_823p	Yes	T62	DQ31	DQ15	DQ7
3A_B	1	IO	IOB		HPS_DDR	DIFF_IO_3A_824n	Yes	G65	DQ31	DQ15	DQ7
3A_B	0	IO	IOB		HPS_DDR	DIFF_IO_3A_824p	Yes	J65	DQ31	DQ15	DQ7
6A		IO	HVIO_6A_1, TXCLK1, Data_Ctrl1					CG26			
6A		IO	HVIO_6A_2, TXCLK2, Data_Ctrl2					CA23			
6A		IO	HVIO_6A_3, TXCLK3, Data_Ctrl3					CG23			
6A		IO	HVIO_6A_4, TXCLK4, Data_Ctrl4					CD23			
6A		IO	HVIO_6A_5, TXCLK5, Data_Ctrl5					CR19			
6A		IO	HVIO_6A_6, TXCLK6, Data_Ctrl6					CV16			
6A		IO	HVIO_6A_7, TXCLK7, Data_Ctrl7					CV19			
6A		IO	HVIO_6A_8, TXCLK8, Data_Ctrl8					CV14			
6A		IO	HVIO_6A_9, PLLREFCLK1, TXCLK9, RXCLK1, Data_Ctrl9					BV14			
6A		IO	HVIO_6A_10, PLLREFCLK2, TXCLK10, RXCLK2, Data_Ctrl10					CE19			
6A		IO	HVIO_6A_11, SOURCE_SYNC_CLK1, TXCLK11, RXCLK3, Data_Ctrl11					BV11			
6A		IO	HVIO_6A_12, SOURCE_SYNC_CLK2, TXCLK12, RXCLK4, Data_Ctrl12					CH19			
6A		IO	HVIO_6A_13, TXCLK13, Data_Ctrl13					CV11			
6A		IO	HVIO_6A_14, TXCLK14, Data_Ctrl14					CR14			
6A		IO	HVIO_6A_15, TXCLK15, Data_Ctrl15					BV16			
6A		IO	HVIO_6A_16, TXCLK16, Data_Ctrl16					CR11			
6A		IO	HVIO_6A_17, TXCLK17, Data_Ctrl17					CH16			
6A		IO	HVIO_6A_18, TXCLK18, Data_Ctrl18					BV19			
6A		IO	HVIO_6A_19, TXCLK19, Data_Ctrl19					CE14			
6A		IO	HVIO_6A_20, TXCLK20, Data_Ctrl20					CH14			
6B		IO	HVIO_6B_1, TXCLK1, Data_Ctrl1					DC2			
6B		IO	HVIO_6B_2, TXCLK2, Data_Ctrl2					DC1			
6B		IO	HVIO_6B_3, TXCLK3, Data_Ctrl3					DE1			
6B		IO	HVIO_6B_4, TXCLK4, Data_Ctrl4					CH4			
6B		IO	HVIO_6B_5, TXCLK5, Data_Ctrl5					DH2			
6B		IO	HVIO_6B_6, TXCLK6, Data_Ctrl6					DH1			
6B		IO	HVIO_6B_7, TXCLK7, Data_Ctrl7					CV6			
6B		IO	HVIO_6B_8, TXCLK8, Data_Ctrl8					CV4			
6B		IO	HVIO_6B_9, PLLREFCLK1, TXCLK9, RXCLK1, Data_Ctrl9					DM2			
6B		IO	HVIO_6B_10, PLLREFCLK2, TXCLK10, RXCLK2, Data_Ctrl10					CH6			
6B		IO	HVIO_6B_11, SOURCE_SYNC_CLK1, TXCLK11, RXCLK3, Data_Ctrl11					CR6			
6B		IO	HVIO_6B_12, SOURCE_SYNC_CLK2, TXCLK12, RXCLK4, Data_Ctrl12					CR8			
6B		IO	HVIO_6B_13, TXCLK13, Data_Ctrl13					CE8			
6B		IO	HVIO_6B_14, TXCLK14, Data_Ctrl14					DO3			
6B		IO	HVIO_6B_15, TXCLK15, Data_Ctrl15					CE11			
6B		IO	HVIO_6B_16, TXCLK16, Data_Ctrl16					DF3			
6B		IO	HVIO_6B_17, TXCLK17, Data_Ctrl17					DN2			
6B		IO	HVIO_6B_18, TXCLK18, Data_Ctrl18					CE6			
6B		IO	HVIO_6B_19, TXCLK19, Data_Ctrl19					CH11			
6B		IO	HVIO_6B_20, TXCLK20, Data_Ctrl20					DI3			
6C		IO	HVIO_6C_1, TXCLK1, Data_Ctrl1					CI9			
6C		IO	HVIO_6C_2, TXCLK2, Data_Ctrl2					AG31			
6C		IO	HVIO_6C_3, TXCLK3, Data_Ctrl3					R19			

Bank Number	Index within I/O Bank	Pin Name/Function	Optional Function(s)	Configuration Function	HPS Function	Dedicated Tx/Rx Channel SERDES	Soft CDR Support	B238	DQS for X4	DQS for X8/X9	DQS for X16/X18
6C		IO	HVIO_6C_4, TXCLK4, Data_Ctrl4					B17			
6C		IO	HVIO_6C_5, TXCLK5, Data_Ctrl5					AL37			
6C		IO	HVIO_6C_6, TXCLK6, Data_Ctrl6					H19			
6C		IO	HVIO_6C_7, TXCLK7, Data_Ctrl7					AL34			
6C		IO	HVIO_6C_8, TXCLK8, Data_Ctrl8					AL31			
6C		IO	HVIO_6C_9, PLLREFCLK1, TXCLK9, RXCLK1, Data_Ctrl9					H16			
6C		IO	HVIO_6C_10, PLLREFCLK2, TXCLK10, RXCLK2, Data_Ctrl10					V16			
6C		IO	HVIO_6C_11, SOURCE_SYNC_CLK1, TXCLK11, RXCLK3, Data_Ctrl11					V19			
6C		IO	HVIO_6C_12, SOURCE_SYNC_CLK2, TXCLK12, RXCLK4, Data_Ctrl12					H14			
6C		IO	HVIO_6C_13, TXCLK13, Data_Ctrl13					AH19			
6C		IO	HVIO_6C_14, TXCLK14, Data_Ctrl14					H11			
6C		IO	HVIO_6C_15, TXCLK15, Data_Ctrl15					AE19			
6C		IO	HVIO_6C_16, TXCLK16, Data_Ctrl16					AH16			
6C		IO	HVIO_6C_17, TXCLK17, Data_Ctrl17					H6			
6C		IO	HVIO_6C_18, TXCLK18, Data_Ctrl18					R14			
6C		IO	HVIO_6C_19, TXCLK19, Data_Ctrl19					V14			
6C		IO	HVIO_6C_20, TXCLK20, Data_Ctrl20					H4			
6D		IO	HVIO_6D_1, TXCLK1, Data_Ctrl1					U2			
6D		IO	HVIO_6D_2, TXCLK2, Data_Ctrl2					Y1			
6D		IO	HVIO_6D_3, TXCLK3, Data_Ctrl3					P1			
6D		IO	HVIO_6D_4, TXCLK4, Data_Ctrl4					P2			
6D		IO	HVIO_6D_5, TXCLK5, Data_Ctrl5					Y2			
6D		IO	HVIO_6D_6, TXCLK6, Data_Ctrl6					L1			
6D		IO	HVIO_6D_7, TXCLK7, Data_Ctrl7					F2			
6D		IO	HVIO_6D_8, TXCLK8, Data_Ctrl8					C6			
6D		IO	HVIO_6D_9, PLLREFCLK1, TXCLK9, RXCLK1, Data_Ctrl9					C2			
6D		IO	HVIO_6D_10, PLLREFCLK2, TXCLK10, RXCLK2, Data_Ctrl10					C8			
6D		IO	HVIO_6D_11, SOURCE_SYNC_CLK1, TXCLK11, RXCLK3, Data_Ctrl11					B3			
6D		IO	HVIO_6D_12, SOURCE_SYNC_CLK2, TXCLK12, RXCLK4, Data_Ctrl12					C11			
6D		IO	HVIO_6D_13, TXCLK13, Data_Ctrl13					C14			
6D		IO	HVIO_6D_14, TXCLK14, Data_Ctrl14					B5			
6D		IO	HVIO_6D_15, TXCLK15, Data_Ctrl15					A10			
6D		IO	HVIO_6D_16, TXCLK16, Data_Ctrl16					B10			
6D		IO	HVIO_6D_17, TXCLK17, Data_Ctrl17					A13			
6D		IO	HVIO_6D_19, TXCLK19, Data_Ctrl19					A15			
6D		IO	HVIO_6D_20, TXCLK20, Data_Ctrl20					B15			
6E		IO	HVIO_6E_1, TXCLK1, Data_Ctrl1					AV16			
6E		IO	HVIO_6E_2, TXCLK2, Data_Ctrl2					AV14			
6E		IO	HVIO_6E_3, TXCLK3, Data_Ctrl3					AT19			
6E		IO	HVIO_6E_4, TXCLK4, Data_Ctrl4					AV11			
6E		IO	HVIO_6E_5, TXCLK5, Data_Ctrl5					AT14			
6E		IO	HVIO_6E_6, TXCLK6, Data_Ctrl6					AV19			
6E		IO	HVIO_6E_7, TXCLK7, Data_Ctrl7					AT11			
6E		IO	HVIO_6E_8, TXCLK8, Data_Ctrl8					BE19			
6E		IO	HVIO_6E_9, PLLREFCLK1, TXCLK9, RXCLK1, Data_Ctrl9					BF23			
6E		IO	HVIO_6E_10, PLLREFCLK2, TXCLK10, RXCLK2, Data_Ctrl10					BH14			
6E		IO	HVIO_6E_11, SOURCE_SYNC_CLK1, TXCLK11, RXCLK3, Data_Ctrl11					BJ23			
6E		IO	HVIO_6E_12, SOURCE_SYNC_CLK2, TXCLK12, RXCLK4, Data_Ctrl12					BH11			
6E		IO	HVIO_6E_13, TXCLK13, Data_Ctrl13					BH16			
6E		IO	HVIO_6E_14, TXCLK14, Data_Ctrl14					BE11			
6E		IO	HVIO_6E_15, TXCLK15, Data_Ctrl15					BE14			
6E		IO	HVIO_6E_16, TXCLK16, Data_Ctrl16					BH19			
6E		IO	HVIO_6E_17, TXCLK17, Data_Ctrl17					BU23			
6E		IO	HVIO_6E_18, TXCLK18, Data_Ctrl18					BR19			
6E		IO	HVIO_6E_19, TXCLK19, Data_Ctrl19					BR11			
6E		IO	HVIO_6E_20, TXCLK20, Data_Ctrl20					BR14			
6F		IO	HVIO_6F_1, TXCLK1, Data_Ctrl1					DA1			
6F		IO	HVIO_6F_2, TXCLK2, Data_Ctrl2					DA2			
6F		IO	HVIO_6F_3, TXCLK3, Data_Ctrl3					CU2			
6F		IO	HVIO_6F_4, TXCLK4, Data_Ctrl4					BV6			
6F		IO	HVIO_6F_5, TXCLK5, Data_Ctrl5					CP1			
6F		IO	HVIO_6F_6, TXCLK6, Data_Ctrl6					CP2			
6F		IO	HVIO_6F_7, TXCLK7, Data_Ctrl7					CM1			
6F		IO	HVIO_6F_8, TXCLK8, Data_Ctrl8					BV4			
6F		IO	HVIO_6F_9, PLLREFCLK1, TXCLK9, RXCLK1, Data_Ctrl9					C11			
6F		IO	HVIO_6F_10, PLLREFCLK2, TXCLK10, RXCLK2, Data_Ctrl10					C12			
6F		IO	HVIO_6F_11, SOURCE_SYNC_CLK1, TXCLK11, RXCLK3, Data_Ctrl11					CF2			
6F		IO	HVIO_6F_12, SOURCE_SYNC_CLK2, TXCLK12, RXCLK4, Data_Ctrl12					CF1			
6F		IO	HVIO_6F_13, TXCLK13, Data_Ctrl13					BR6			
6F		IO	HVIO_6F_14, TXCLK14, Data_Ctrl14					CB2			
6F		IO	HVIO_6F_15, TXCLK15, Data_Ctrl15					BW1			
6F		IO	HVIO_6F_16, TXCLK16, Data_Ctrl16					BW2			
6F		IO	HVIO_6F_17, TXCLK17, Data_Ctrl17					BT1			
6F		IO	HVIO_6F_18, TXCLK18, Data_Ctrl18					BR8			
6F		IO	HVIO_6F_19, TXCLK19, Data_Ctrl19					BP1			
6F		IO	HVIO_6F_20, TXCLK20, Data_Ctrl20					BP2			
6G		IO	HVIO_6G_1, TXCLK1, Data_Ctrl1					R11			
6G		IO	HVIO_6G_2, TXCLK2, Data_Ctrl2					AH14			
6G		IO	HVIO_6G_3, TXCLK3, Data_Ctrl3					V11			

Bank Number	Index within I/O Bank	Pin Name/Function	Optional Function(s)	Configuration Function	HPS Function	Dedicated Tx/Rx Channel SERDES	Soft CDR Support	B238	DQS for X4	DQS for X8/X9	DQS for X16/X18
EG		IO	HVIO_6G_4, TXCLK4, Data, Ctrl4					R8			
EG		IO	HVIO_6G_5, TXCLK5, Data, Ctrl5					V6			
EG		IO	HVIO_6G_6, TXCLK6, Data, Ctrl6					AE14			
EG		IO	HVIO_6G_7, TXCLK7, Data, Ctrl7					R6			
EG		IO	HVIO_6G_8, TXCLK8, Data, Ctrl8					V4			
EG		IO	HVIO_6G_9, PLLREFCLK1, TXCLK9, RXCLK1, Data, Ctrl9					AE11			
EG		IO	HVIO_6G_10, PLLREFCLK2, TXCLK10, RXCLK2, Data, Ctrl10					AG23			
EG		IO	HVIO_6G_11, SOURCE_SYNC_CLK1, TXCLK11, RXCLK3, Data, Ctrl11					AH11			
EG		IO	HVIO_6G_12, SOURCE_SYNC_CLK2, TXCLK12, RXCLK4, Data, Ctrl12					AE8			
EG		IO	HVIO_6G_13, TXCLK13, Data, Ctrl13					AG26			
EG		IO	HVIO_6G_14, TXCLK14, Data, Ctrl14					AE6			
EG		IO	HVIO_6G_15, TXCLK15, Data, Ctrl15					AH6			
EG		IO	HVIO_6G_16, TXCLK16, Data, Ctrl16					AH4			
EG		IO	HVIO_6G_17, TXCLK17, Data, Ctrl17					AT8			
EG		IO	HVIO_6G_18, TXCLK18, Data, Ctrl18					AV4			
EG		IO	HVIO_6G_19, TXCLK19, Data, Ctrl19					AV6			
EG		IO	HVIO_6G_20, TXCLK20, Data, Ctrl20					AT6			
EH		IO	HVIO_6H_1, TXCLK1, Data, Ctrl1					BL2			
EH		IO	HVIO_6H_2, TXCLK2, Data, Ctrl2					BL1			
EH		IO	HVIO_6H_3, TXCLK3, Data, Ctrl3					BH4			
EH		IO	HVIO_6H_4, TXCLK4, Data, Ctrl4					BG2			
EH		IO	HVIO_6H_5, TXCLK5, Data, Ctrl5					BD2			
EH		IO	HVIO_6H_6, TXCLK6, Data, Ctrl6					BD1			
EH		IO	HVIO_6H_7, TXCLK7, Data, Ctrl7					BH6			
EH		IO	HVIO_6H_8, TXCLK8, Data, Ctrl8					BA1			
EH		IO	HVIO_6H_9, PLLREFCLK1, TXCLK9, RXCLK1, Data, Ctrl9					AU2			
EH		IO	HVIO_6H_10, PLLREFCLK2, TXCLK10, RXCLK2, Data, Ctrl10					AU1			
EH		IO	HVIO_6H_11, SOURCE_SYNC_CLK1, TXCLK11, RXCLK3, Data, Ctrl11					AP1			
EH		IO	HVIO_6H_12, SOURCE_SYNC_CLK2, TXCLK12, RXCLK4, Data, Ctrl12					AP2			
EH		IO	HVIO_6H_13, TXCLK13, Data, Ctrl13					AM2			
EH		IO	HVIO_6H_14, TXCLK14, Data, Ctrl14					BE6			
EH		IO	HVIO_6H_15, TXCLK15, Data, Ctrl15					AJ2			
EH		IO	HVIO_6H_16, TXCLK16, Data, Ctrl16					AJ1			
EH		IO	HVIO_6H_17, TXCLK17, Data, Ctrl17					BE8			
EH		IO	HVIO_6H_18, TXCLK18, Data, Ctrl18					AF1			
EH		IO	HVIO_6H_19, TXCLK19, Data, Ctrl19					AC2			
EH		IO	HVIO_6H_20, TXCLK20, Data, Ctrl20					AC1			
		GND						CU74			
		GND						BM67			
		GND						BM64			
		GND						AL54			
		GND						AL52			
		GND						AW57			
		GND						AR57			
		GND						VW65			
		GND						VW59			
		GND						VW51			
		GND						VW42			
		GND						VW35			
		GND						VW27			
		GND						V8			
		GND						U75			
		GND						U1			
		GND						R4			
		GND						R16			
		GND						L74			
		GND						L2			
		GND						K72			
		GND						J62			
		GND						J56			
		GND						J48			
		GND						J39			
		GND						J32			
		GND						J24			
		GND						H8			
		GND						F1			
		GND						E65			
		GND						E59			
		GND						E51			
		GND						E42			
		GND						E35			
		GND						E27			
		GND						DP74			
		GND						DP73			
		GND						DP63			
		GND						DP53			
		GND						DP41			
		GND						DP3			
		GND						DP28			

Bank Number	Index within I/O Bank	Pin Name/Function	Optional Function(s)	Configuration Function	HPS Function	Dedicated Tx/Rx Channel SERDES	Soft CDR Support	B238	DQS for X4	DQS for X8/X9	DQS for X16/X18
		GND						DP2			
		GND						DP17			
		GND						DN75			
		GND						DN70			
		GND						DN60			
		GND						DN47			
		GND						DN36			
		GND						DN23			
		GND						DN13			
		GND						DN1			
		GND						DM75			
		GND						DM74			
		GND						DM1			
		GND						DL72			
		GND						DK65			
		GND						DK59			
		GND						DK51			
		GND						DK42			
		GND						DK35			
		GND						DK3			
		GND						DK27			
		GND						DK21			
		GND						DK12			
		GND						DG72			
		GND						DF9			
		GND						DF62			
		GND						DF56			
		GND						DF48			
		GND						DF39			
		GND						DF32			
		GND						DF24			
		GND						DF18			
		GND						DE74			
		GND						DE2			
		GND						D72			
		GND						CV72			
		GND						CV64			
		GND						CW54			
		GND						CW46			
		GND						CW37			
		GND						CW29			
		GND						CV8			
		GND						CU75			
		GND						CU1			
		GND						CT57			
		GND						CT49			
		GND						CT40			
		GND						CT31			
		GND						CR4			
		GND						CR16			
		GND						CN69			
		GND						CM74			
		GND						CM2			
		GND						CL60			
		GND						CL52			
		GND						CL44			
		GND						CL34			
		GND						CL26			
		GND						CL23			
		GND						CK72			
		GND						CK64			
		GND						CH8			
		GND						CG54			
		GND						CG46			
		GND						CG37			
		GND						CG29			
		GND						CE4			
		GND						CE16			
		GND						CD57			
		GND						CD49			
		GND						CD40			
		GND						CD31			
		GND						CC69			
		GND						CB75			
		GND						CB1			
		GND						CA60			
		GND						CA52			
		GND						CA44			
		GND						CA34			

Bank Number	Index within I/O Bank	Pin Name/Function	Optional Function(s)	Configuration Function	HPS Function	Dedicated Tx/Rx Channel SERDES	Soft CDR Support	B238	DQS for X4	DQS for X8/X9	DQS for X16/X18
		GND						CA26			
		GND						C75			
		GND						C4			
		GND						C16			
		GND						C1			
		GND						BY72			
		GND						BY64			
		GND						BV8			
		GND						BU60			
		GND						BU57			
		GND						BU54			
		GND						BU46			
		GND						BU37			
		GND						BU29			
		GND						BT74			
		GND						BT2			
		GND						BR4			
		GND						BR16			
		GND						BN57			
		GND						BN49			
		GND						BN40			
		GND						BN31			
		GND						BN23			
		GND						BM69			
		GND						BK72			
		GND						BK64			
		GND						BJ57			
		GND						BJ52			
		GND						BJ44			
		GND						BJ34			
		GND						BH8			
		GND						BG75			
		GND						BG1			
		GND						BF60			
		GND						BF54			
		GND						BF46			
		GND						BF37			
		GND						BF29			
		GND						BE4			
		GND						BE16			
		GND						BC69			
		GND						BB60			
		GND						BB57			
		GND						BB49			
		GND						BB40			
		GND						BB31			
		GND						BB23			
		GND						BA74			
		GND						BA2			
		GND						B75			
		GND						B70			
		GND						B60			
		GND						B47			
		GND						B36			
		GND						B23			
		GND						B20			
		GND						B13			
		GND						AY72			
		GND						AY64			
		GND						AW60			
		GND						AW54			
		GND						AW52			
		GND						AW44			
		GND						AW37			
		GND						AW26			
		GND						AV8			
		GND						AT4			
		GND						AT16			
		GND						AR60			
		GND						AR54			
		GND						AR46			
		GND						AR44			
		GND						AR29			
		GND						AN69			
		GND						AM75			
		GND						AM1			
		GND						AL60			
		GND						AL57			
		GND						AL49			

Bank Number	Index within I/O Bank	Pin Name/Function	Optional Function(s)	Configuration Function	HPS Function	Dedicated Tx/Rx Channel SERDES	Soft CDR Support	B238	DQS for X4	DQS for X8/X9	DQS for X16/X18
		GND						AL44			
		GND						AL40			
		GND						AL26			
		GND						AL23			
		GND						AK72			
		GND						AK64			
		GND						AH8			
		GND						AG60			
		GND						AG57			
		GND						AG54			
		GND						AG52			
		GND						AG49			
		GND						AG44			
		GND						AG34			
		GND						AF74			
		GND						AF2			
		GND						AE4			
		GND						AE16			
		GND						AB65			
		GND						AB59			
		GND						AB51			
		GND						AB42			
		GND						AB35			
		GND						AB27			
		GND						AA72			
		GND						A74			
		GND						A73			
		GND						A63			
		GND						A53			
		GND						A5			
		GND						A41			
		GND						A3			
		GND						A28			
		GND						A17			
		GNDSENSE						BN44			
		VCC						CW44			
		VCC						CW40			
		VCC						CT44			
		VCC						CL40			
		VCC						CG44			
		VCC						CG40			
		VCC						CD46			
		VCC						CD44			
		VCC						CA40			
		VCC						CA37			
		VCC						BU49			
		VCC						BU44			
		VCC						BU40			
		VCC						BU34			
		VCC						BN37			
		VCC						BN34			
		VCC						BJ49			
		VCC						BJ46			
		VCC						BJ40			
		VCC						BF49			
		VCC						BF44			
		VCCPT						CL57			
		VCCPT						CL54			
		VCCPT						AG40			
		VCCPT						AG37			
		VCCPT_HVIO						CW60			
		VCCPT_HVIO						CT60			
		VCCPT_HVIO						CW34			
		VCCPT_HVIO						CW31			
		VCCPT_HVIO						CA31			
		VCCPT_HVIO						BU31			
		VCCPT_HVIO						BJ31			
		VCCPT_HVIO						BF31			
		VCCPT_HVIO						BB29			
		VCCPT_HVIO						BB26			
		DNU						CC71			
		DNU						BM72			
		DNU						CP75			
		DNU						CP74			
		DNU						BJ60			
		DNU						BN60			
		DNU						CT34			
		DNU						AR34			
		DNU						CG57			

Bank Number	Index within I/O Bank	Pin Name/Function	Optional Function(s)	Configuration Function	HPS Function	Dedicated Tx/Rx Channel SERDES	Soft CDR Support	B238	DQS for X4	DQS for X8/X9	DQS for X16/X18
		DNU						CD26			
		DNU						B2			
		DNU						AR31			
		DNU						CT52			
		DNU						AR26			
		DNU						CG60			
		DNU						CT23			
		DNU						AW23			
		DNU						CD60			
		DNU						CW23			
		DNU						AR23			
		VCCBAT						CG49			
		VCCIO_HPS						BB46			
		VCCIO_HPS						BB44			
		VCCIO_HVIO_6A						CT29			
		VCCIO_HVIO_6A						CL29			
		VCCIO_HVIO_6B						CW26			
		VCCIO_HVIO_6B						CT26			
		VCCIO_HVIO_6C						AL29			
		VCCIO_HVIO_6C						AG29			
		VCCIO_HVIO_6D						AW31			
		VCCIO_HVIO_6D						AW29			
		VCCIO_HVIO_6E						CD29			
		VCCIO_HVIO_6E						CA29			
		VCCIO_HVIO_6F						BU26			
		VCCIO_HVIO_6F						BN26			
		VCCIO_HVIO_6G						BN29			
		VCCIO_HVIO_6G						BI29			
		VCCIO_HVIO_6H						BI26			
		VCCIO_HVIO_6H						BF26			
		VCCIO_PIO_2A_B						CT37			
		VCCIO_PIO_2A_B						CL37			
		VCCIO_PIO_2A_T						CL31			
		VCCIO_PIO_2A_T						CG31			
		VCCIO_PIO_3A_B						AW40			
		VCCIO_PIO_3A_B						AR40			
		VCCIO_PIO_3A_T						AW34			
		VCCIO_PIO_3A_T						AR37			
		VCCIO_PIO_SDM						CW57			
		VCCIO_SDM						CW49			
		NC						DL71			
		NC						DH74			
		NC						DE75			
		NC						CT46			
		NC						CM75			
		NC						CC72			
		NC						CB74			
		NC						BT75			
		NC						B74			
		NC						B73			
		VCCRCORE						BU52			
		VCCRCORE						BN52			
		VCCRCORE						BJ37			
		VCCADC						CG52			
		VCCFUSEWR_SDM						CD52			
		VCCH_SDM						CD54			
		VCCLSENSE						BN46			
		VCCL_ADC_SDM						CW52			
		VCCL_HPS						AW49			
		VCCL_HPS						AW46			
		VCCL_HPS						AR52			
		VCCL_HPS						AR49			
		VCCL_HPS_CORE0_CORE1						AL46			
		VCCL_HPS_CORE0_CORE1						AG46			
		VCCL_SDM						CA57			
		VCCL_SDM						CA54			
		VCCL_SDM						CA49			
		VCCL_SDM						CA46			
		VCCP						CG34			
		VCCP						CD37			
		VCCP						CD34			
		VCCP						BF40			
		VCCP						BF34			
		VCCP						BB37			
		VCCP						BB34			
		VCCPLL1_HPS						BF57			
		VCCPLL2_HPS						BF52			
		VCCPLLDIG1_HPS						BB54			
		VCCPLLDIG2_HPS						BB52			

Date	Version	Revision History for B23B Package
July 2024	2024-07-08	Initial release.
January 2025	2025-01-13	Added I3C0_SDA_PULLUP_EN and I3C1_SDA_PULLUP_EN signal.
April 2025	2025-04-07	-Updated B2 pin in bank 6D to DNU. -Updated total HVIO to 159.
June 2025	2025-06-13	Updated soft CDR support column: -Added CDR support on pin index 0, 1, 12, 13, 24, 25, 36, and 37. -Removed CDR support on pin index 10, 11, 22, 23, 34, 35, 46, and 47.

(1) For more information about pin definition and pin connection guidelines, refer to the

[Pin Connection Guidelines: Agilex™ 5 FPGAs and SoCs](#)

(2) IOB for IO pin name/function is equivalent to HSIO.

(3) To obtain a Comma Delimited (CSV) file using Excel - From the active tab, select "Save As" in the FILE menu and then select CSV (*.csv) format and SAVE.

CSV File may be viewed in a TEXT editor or in Excel.